Songshi Dou

Ph.D. Candidate
Department of Electrical and Computer Engineering (ECE)
The University of Hong Kong (HKU)
Room 501, Chow Yei Ching Bldg., Pokfulam Road, Hong Kong

Personal Email: songshidou@hotmail.com
Work Email: ssdou@eee.hku.hk
`https://songshidou.github.io`
(+852) 9748 2397

Summary / Statement

Songshi Dou is a fourth-year Ph.D. candidate at the HKU. His research mainly focuses on computer and communication networking, including satellite Internet, software-defined networking (SDN), and traffic engineering (TE). He has published 44 peer-reviewed research papers in leading venues, including JSAC, ToN, TSC, TCOM, WWW, INFOCOM, ICDCS, and IWQoS, and holds 5 Chinese patents

Education

The University of Hong Kong (HKU), Pokfulam, Hong Kong — 2023 - 2027 (expc.)

Ph.D. Candidate in Electrical and Computer Engineering
Field of Study: Computer and Communication Networking
Supervisors: Prof. Lawrence K. Yeung and Prof. Xianhao Chen

Beijing Institute of Technology (BIT), Beijing, China — 2019 - 2022

M.E. in Control Engineering, June 2022
Field of Study: Computer Networking
Supervisor: Prof. Zehua Guo
Dissertation: Maintaining the Path Programmability in Software-Defined Wide Area Networks
Outstanding Master's Thesis of Chinese Institute of Electronics

North China Electric Power University (NCEPU), Beijing, China — 2015 - 2019

B.E. in Automation, June 2019

Publications

(†Equal contribution, *Corresponding author)

Journal & Magazine Papers

[J25] Chong Wu, Zehua Guo, and **Songshi Dou**, "A Little Data is Enough: Criticality-Aware Range Routing for Wide

Area Networks", *IEEE Transactions on Networking* (**ToN**), Accepted.

[J24] **Songshi Dou**, Feihu Jin, Jinxian Wu, A-Long Jin, and Lawrence K. Yeung, "SpaceMeet: Bringing Conferencing Closer through In-Orbit Conferencing Services", *IEEE Transactions on Services Computing* (**TSC**), Accepted.

[J23] **Songshi Dou** and Zehua Guo, "Quest: Quality of Service-centric Resilient Routing for Software-Defined Wide Area Networks", *IEEE Transactions on Networking* (**ToN**), vol. 34, pp. 4555-4569, 2026.

[J22] **Songshi Dou**, Shengyu Zhang, Xianhao Chen, Zehua Guo, Tony Q.S. Quek, and Lawrence K. Yeung, "Beyond Space Routers: Revisiting the Role of LEO Satellite Constellations in Future Global Internet Architecture", *IEEE Wireless Communications* (**WCM**), vol. 33, no. 3, pp. 31-38, 2026.

[J21] Shengyu Zhang, **Songshi Dou***, Zhenglong Li, Lawrence K. Yeung, and Tony Q.S. Quek, "Maintaining Predictable QoS for Online Service Provisioning in Non-Terrestrial Networks via Safe Transfer Learning", *IEEE Transactions on Networking* (**ToN**), vol. 34, pp. 1745-1760, 2026.

[J20] **Songshi Dou**, Jinxian Wu, Shengyu Zhang, Xianhao Chen, Tony Q.S. Quek, and Lawrence K. Yeung, "Matchmaker: Maintaining QoS-aware and Predictable Load Balancing Performance for LEO Mega-Constellations", *IEEE Transactions on Communications* (**TCOM**), vol. 73, no. 12, pp. 14078-14092, 2025.

[J19] **Songshi Dou**, Shengyu Zhang, Zhenglong Li, Jinxian Wu, Xianhao Chen, and Lawrence K. Yeung, "SpaceCache+: Towards Pervasive Content Delivery via Low-Earth Orbit Mega-Constellations", *IEEE Transactions on Services Computing* (**TSC**), vol. 18, no. 5, pp. 3206-3220, 2025.

[J18] **Songshi Dou**, Zehua Guo, and Lawrence K. Yeung, "Unleashing the Potential of LEO Constellations in Building Resilient and Low-Latency Control Plane for SD-WANs", *IEEE Transactions on Networking* (**ToN**), vol. 33, no. 5, pp. 2706-2719, 2025.

[J17] Jinxian Wu, Li Dai, **Songshi Dou**, Yunshan Deng, and Yuanqing Xia, "Towards Improved Performance of Inner Convex Approximation for Suboptimal Nonlinear MPC", *IEEE Transactions on Cybernetics* (**TCYB**), vol. 55, no. 9, pp. 4322-4332, 2025.

[J16] Jinxian Wu, Li Dai, **Songshi Dou**, and Yuanqing Xia, "Accelerated Successive Convex Approximation for Nonlinear Optimization-Based Control", *IEEE Transactions on Automatic Control* (**TAC**), vol. 70, no. 9, pp. 6237-6244, 2025.

[J15] **Songshi Dou** and Zehua Guo, "Maintaining Predictable Traffic Engineering Performance under Controller Failures for Software-Defined WANs", *IEEE Journal on Selected Areas in Communications* (**JSAC**), vol. 43, no. 2, pp. 524-536, 2025.

[J14] Tengteng Zhu, Zehua Guo, Chao Yao, Jiaxin Tan, **Songshi Dou**, Wenrun Wang, and Zhenzhen Han, "Byzantine-robust Federated Learning via Cosine Similarity Aggregation", *Elsevier Computer Networks* (**COMNET**), vol. 254, p. 110730, 2024.

[J13] **Songshi Dou**, Li Qi, Jianye Wang, and Zehua Guo, "EPIC: Traffic Engineering-centric Path Programmability Recovery under Controller Failures in SD-WANs", *IEEE/ACM Transactions on Networking* (**ToN**), vol. 32, no. 6, pp. 4871-4884, 2024.

[J12] Zehua Guo, Li Qi, **Songshi Dou**, Jiawei Weng, Xiaoyang Fu, and Yuanqing Xia, "Maintaining Control Resiliency for Traffic Engineering in SD-WANs", *IEEE/ACM Transactions on Networking* (**ToN**), vol. 32, no. 4, pp. 3485-3498, 2024.

[J11] **Songshi Dou** and Zehua Guo, "Path Programmability Recovery under Controller Failures for SD-WANs: Recent Advances and Future Research Challenges", *IEEE Communications Magazine* (**COMMAG**), vol. 62, no. 11, pp. 100-106, 2024.

[J10] **Songshi Dou**, Li Qi, and Zehua Guo, "Mitigating the Impact of Controller Failures on QoS Robustness for Software-Defined Wide Area Networks", *Elsevier Computer Networks* (**COMNET**), vol. 238, p. 110096, 2024.

[J9] Zehua Guo, Changlin Li, Yang Li, **Songshi Dou**, Bida Zhang, and Weichao Wu, "Maintaining the Network Performance of Software-Defined WANs with Efficient Critical Routing", *IEEE Transactions on Network and Service Management* (**TNSM**), vol. 21, no. 2, pp. 2240-2252, 2024.

[J8] Yuntian Zhang, Ning Han, Tengteng Zhu, Junjie Zhang, Minghao Ye, **Songshi Dou**, and Zehua Guo, "Prophet: Traffic Engineering-centric Traffic Matrix Prediction", *IEEE/ACM Transactions on Networking* (**ToN**), vol. 32, no. 1, pp. 822-832, 2024.

[J7] Zehua Guo, **Songshi Dou**, Wenchao Jiang, and Yuanqing Xia, "Toward Improved Path Programmability Recovery for Software-Defined WANs under Multiple Controller Failures", *IEEE/ACM Transactions on Networking* (**ToN**), vol. 32, no. 1, pp. 143-158, 2024.

[J6] **Songshi Dou**†, Li Qi†, Chao Yao, and Zehua Guo, "Exploring the Impact of Critical Programmability on Controller Placement for Software-Defined Wide Area Networks", *IEEE/ACM Transactions on Networking* (**ToN**), vol. 31, no. 6, pp. 2575-2588, 2023.

[J5] Zehua Guo, **Songshi Dou***, Wenfei Wu, and Yuanqing Xia, "Toward Flexible and Predictable Path Programmability Recovery under Multiple Controller Failures in Software-Defined WANs", *IEEE/ACM Transactions on Networking* (**ToN**), vol. 31, no. 5, pp. 1965-1980, 2023.

[J4] Zehua Guo, **Songshi Dou**, Sen Liu, Wendi Feng, Wenchao Jiang, Yang Xu, and Zhi-Li Zhang, "Maintaining Control Resiliency and Flow Programmability in Software-Defined WANs During Controller Failures", *IEEE/ACM Transactions on Networking* (**ToN**), vol. 30, no. 3, pp. 969-984, 2022.

[J3] Haoran Ni, Zehua Guo, Changlin Li, **Songshi Dou**, Chao Yao, and Thar Baker, "Network Coding-based Resilient Routing for Maintaining Data Security and Availability in Software-Defined Networks", *Elsevier Journal of Network and Computer Applications* (**JNCA**), vol. 205, p. 103372, 2022.

[J2] Zehua Guo, **Songshi Dou**, Yi Wang, Sen Liu, Wendi Feng, and Yang Xu, "HybridFlow: Achieving Load Balancing in Software-Defined WANs with Scalable Routing", *IEEE Transactions on Communications* (**TCOM**), vol. 69, no. 8, pp. 5255-5268, 2021.

[J1] **Songshi Dou**, Guochun Miao, Zehua Guo, Chao Yao, Weiran Wu, and Yuanqing Xia, "Matchmaker: Maintaining Network Programmability for Software-Defined WANs under Multiple Controller Failures", *Elsevier Computer Networks* (**COMNET**), vol. 192, p. 108045, 2021.

Conference Papers

[C13] Xiaoyang Fu, Zehua Guo, and **Songshi Dou**, "QoS-driven Network Soft Slicing through Token-based Hierarchical Scheduling", *IEEE/ACM International Symposium on Quality of Service 2026* (**IWQoS'26**).

[C12] Chong Wu, Zehua Guo, **Songshi Dou**, and Minghao Ye, "Ensuring QoS Stability in Dynamic WANs through Selectively Optimized Range Routing", *IEEE/ACM International Symposium on Quality of Service 2026* (**IWQoS'26**).

[C11] Jixing Yang, Shuran Zhang, Zehua Guo, **Songshi Dou**, and Jianye Wang, "CEINR: Critical Element-aware In-Network Recovery for Distributed Machine Learning", *IWQoS Workshop on Generative Cloud-Network Infrastructure 2026* (**IWQoS GCNI'26**).

[C10] **Songshi Dou**, Jinxian Wu, A-Long Jin, Zehua Guo, and Lawrence K. Yeung, "Rethinking Nearest-Is-Best: Toward Load-aware Ground Station Assignment for LEO Satellite Networks", *IEEE/IFIP Network Operations and Management Symposium 2026* (**NOMS'26**).

[C9] Shengyu Zhang, **Songshi Dou***, Zhenglong Li, Lawrence K. Yeung, and Tony Q.S. Quek, "ORACLE: QoS-aware Online Service Provisioning in Non-Terrestrial Networks with Safe Transfer Learning", *IEEE International Conference on Computer Communications 2025* (**INFOCOM'25**). (**Accept Ratio: 272/1458=18.66%**)

[C8] Junwei Wu, **Songshi Dou***, and Lawrence K. Yeung, "Entanglement-Reusable Dynamic Routing for Quantum Networks", *IEEE International Conference on Consumer Electronics 2025* (**ICCE'25**). (**Best Paper Candidate Award**)

[C7] **Songshi Dou**, Xianhao Chen, and Lawrence K. Yeung, "Enabling Practical and Pervasive Content Delivery from Emerging LEO Mega-Constellations", *IEEE International Conference on Multimedia and Expo 2024* (**ICME'24**).

[C6] **Songshi Dou**, Li Qi, and Zehua Guo, "ARES: Predictable Traffic Engineering under Controller Failures in SD-WANs", *ACM Web Conference 2024* (**WWW'24**). (**Accept Ratio: $\sim$20.2%**)

[C5] **Songshi Dou**, Shengyu Zhang, and Lawrence K. Yeung, "Achieving Predictable and Scalable Load Balancing Performance in LEO Mega-Constellations", *IEEE International Conference on Communications 2024* (**ICC'24**).

[C4] **Songshi Dou**, Yongchao He, Sen Liu, Wenfei Wu, and Zehua Guo, "RateSheriff: Multipath Flow-aware and Resource Efficient Rate Limiter Placement for Data Center Networks", *IEEE/ACM International Symposium on Quality of Service 2023* (**IWQoS'23**). (**Accept Ratio: 62/264=23.5%**)

[C3] **Songshi Dou**, Zehua Guo, and Yuanqing Xia, "ProgrammabilityMedic: Predictable Path Programmability Recovery under Multiple Controller Failures in SD-WANs", *IEEE International Conference on Distributed Computing Systems 2021* (**ICDCS'21**). (**Accept Ratio: 97/489=19.8%**)

[C2] Yijun Sun, Zehua Guo, **Songshi Dou**, and Yuanqing Xia, "Video Quality and Popularity-aware Video Caching in Content Delivery Networks", *IEEE International Conference on Web Services 2021* (**ICWS'21**).

[C1] Zehua Guo, **Songshi Dou**, and Wenchao Jiang, "Improving the Path Programmability for Software-Defined WANs under Multiple Controller Failures", *IEEE/ACM International Symposium on Quality of Service 2020* (**IWQoS'20**).

Workshop Papers

[W2] Jixing Yang, Shuran Zhang, Zehua Guo, **Songshi Dou**, and Jianye Wang, "CEINR: Critical Element-aware In-Network Recovery for Distributed Machine Learning", *IWQoS Workshop on Generative Cloud-Network Infrastructure 2026* (**IWQoS GCNI'26**).

[W1] Li Qi†, **Songshi Dou**†, Zehua Guo, Changlin Li, Yang Li, and Tengteng Zhu, "Low Control Latency SD-WANs for Metaverse", *International Workshop on Social and Metaverse Computing and Networking 2022* (**SocialMeta'22**).

Posters & Demos

[D3] **Songshi Dou**, Jinxian Wu, Zehua Guo, and Lawrence K. Yeung, "Load-aware Ground Station Assignment for LEO Satellite Networks", *IEEE Consumer Communications and Networking Conference 2026* (**CCNC'26**).

[D2] **Songshi Dou**, Li Qi, and Zehua Guo, "Maintaining QoS-aware and Resilient Path Programmability for Metaverse in SD-WANs", *ACM Turing Award Celebration Conference 2023* (**TURC'23**).

[D1] Yijun Sun, Zehua Guo, **Songshi Dou**, Junjie Zhang, Changlin Li, and Xiang Ouyang, "Poster: Enabling Fast Forwarding in Hybrid Software-Defined Networks", *IEEE International Conference on Network Protocols 2021* (**ICNP'21**).

Chinese Publications (in Chinese)

[A2] Yazhu Zhao, Zehua Guo, **Songshi Dou**, and Xiaoyang Fu, "Recent Advances in Programmable Schedulers", *Journal of Electronics and Information Technology* (**JEIT**), 47(12): 1-12, 2025.

[A1] Zehua Guo, **Songshi Dou**, Li Qi, and Julong Lan, "A Survey of Maintaining the Path Programmability in Software-Defined Wide Area Networks", *Journal of Electronics and Information Technology* (**JEIT**), 45(5): 1899-1910, 2023.

Patents

[P5] Zehua Guo, Yutian Zhang, Ning Han, and **Songshi Dou**, "一种以流量工程为中心的流量矩阵预测方法", Chinese Patent, ZL202110810615.0.

[P4] Zehua Guo, **Songshi Dou**, and Yuanqing Xia "一种软件定义广域网中实现负载均衡的可扩展路由方法", Chinese Patent, ZL202010974299.6.

[P3] Zehua Guo, and **Songshi Dou**, "软件定义网络中多控制器失效时流的可编程性优化方法", Chinese Patent, ZL202010544094.4.

[P2] Zehua Guo, Penghao Sun, **Songshi Dou**, Yutian Zhang, Ning Han, and Yuanqing Xia, "基于强化学习的数据中心网络能耗和服务质量优化方法", Chinese Patent, ZL202010308862.6.

[P1] Zehua Guo, Penghao Sun, **Songshi Dou**, Yuanqing Xia, and Honghai Ji, "一种在软件定义网络中实现控制器负载均衡的方法", Chinese Patent, ZL202010094237.6.

Academic Services

Technical Program Committee Member

- IEEE International Parallel and Distributed Processing Symposium (**IPDPS**) *2026*

Reviewer for Journals

- IEEE Journal on Selected Areas in Communications (**JSAC**)
- IEEE/ACM Transactions on Networking (**ToN**)
- IEEE Transactions on Mobile Computing (**TMC**)
- IEEE Transactions on Services Computing (**TSC**)
- IEEE Transactions on Information Forensics and Security (**TIFS**)
- IEEE Transactions on Dependable and Secure Computing (**TDSC**)
- IEEE Transactions on Image Processing (**TIP**)
- IEEE Transactions on Wireless Communications (**TWC**)
- IEEE Transactions on Communications (**TCOM**)

- IEEE Transactions on Network Science and Engineering (**TNSE**)
- IEEE Transactions on Network and Service Management (**TNSM**)
- IEEE Transactions on Vehicular Technology (**TVT**)
- IEEE Transactions on Industrial Informatics (**TII**)
- IEEE Transactions on Machine Learning in Communications and Networking (**TMLCN**)
- IEEE Transactions on Green Communications and Networking (**TGCN**)
- IEEE Transactions on Automatic Control (**TAC**)
- IEEE Transactions on Control of Network Systems (**TCNS**)
- IEEE Transactions on Cybernetics (**TCYB**)
- IEEE Transactions on Aerospace and Electronic Systems (**TAES**)
- IEEE Transactions on Emerging Topics in Computational Intelligence (**TETCI**)
- IEEE Transactions on Reliability (**TR**)
- IEEE Wireless Communications (**WCM**)
- IEEE Network
- IEEE Internet of Things Journal (**IoTJ**)
- IEEE Systems Journal (**ISJ**)
- IEEE Open Journal of the Communications Society (**OJCOMS**)
- IEEE Wireless Communications Letters (**WCL**)
- IEEE Networking Letters (**NL**)
- Science China Information Sciences (**SCIS**)
- Elsevier Journal of Network and Computer Applications (**JNCA**)
- Elsevier Computer Networks (**COMNET**)
- Elsevier Future Generation Computer Systems (**FGCS**)
- Elsevier Ad Hoc Networks
- Elsevier Knowledge-Based Systems (**KBS**)
- EURASIP Journal on Wireless Communications and Networking (**JWCN**)

Reviewer for Conferences

- ACM Multimedia (**MM**) *2024 - 2026*
- IEEE International Conference on Multimedia and Expo (**ICME**) *2024 - 2026*
- IEEE International Conference on Communications (**ICC**) *2023 - 2026*
- IEEE Global Communications Conference (**GLOBECOM**) *2022, 2024 - 2026*
- IEEE International Conference on Computer Communications and Networks (**ICCCN**) *2026*
- IEEE International Symposium on Parallel and Distributed Processing with Applications (**ISPA**) *2024 - 2025*
- IEEE Conference on Decision and Control (**CDC**) *2025*
- IEEE International Conference on Distributed Computing Systems (**ICDCS**) *2024*
- IEEE/ACM International Symposium on Quality of Service (**IWQoS**) *2024*
- International Conference on Knowledge Science, Engineering and Management (**KSEM**) *2022 - 2023*
- International Teletraffic Congress (**ITC**) *2022*

Teaching Experience

- **Teaching Assistant**, IP Networks (ELEC6097) *Fall 2024, Fall 2025, Fall 2026*
- **Tutorial Teacher**, Everyday Computing and the Internet (CCST9003) *Fall 2023*

Honors & Awards

- **1st Runner-Up Award**, IET Young Professionals Exhibition & Competition (YPEC) *2025*
- **Best Paper Candidate Award**, IEEE International Conference on Consumer Electronics (ICCE) *2025*
- **Postgraduate Scholarships**, The University of Hong Kong *2023-2027*
- **Certificate in Teaching and Learning in Higher Education**, The University of Hong Kong *2023*
- **Outstanding Master's Thesis**, Chinese Institute of Electronics *2022*
- **Outstanding Master's Thesis**, Beijing Institute of Technology *2022*
- **Outstanding Graduates**, Beijing Institute of Technology *2022*
- **National Scholarship Award (Top 1%)**, Chinese Ministry of Education *2021*
- **Student Grant**, USENIX Symposium on Operating Systems Design and Implementation (OSDI) *2021*
- **Student Registration Award**, IEEE International Conference on Network Protocols (ICNP) *2021*

- **Student Registration Award**, IEEE International Conference on Distributed Computing Systems (ICDCS)*2021*

Presentations

- "Rethinking Nearest-Is-Best: Toward Load-aware Ground Station Assignment for LEO Satellite Networks", *IEEE/IFIP Network Operations and Management Symposium 2026* (**NOMS'26**), Rome, Italy, May 2026.
- "Load-aware Ground Station Assignment for LEO Satellite Networks", *IEEE Consumer Communications and Networking Conference 2026* (**CCNC'26**), Las Vegas, NV, USA, January 2026.
- "Oracle: QoS-aware Online Service Provisioning in Non-Terrestrial Networks with Safe Transfer Learning", *IEEE International Conference on Computer Communications 2025* (**INFOCOM'25**), London, UK, May 2025.
- "Entanglement-Reusable Dynamic Routing for Quantum Networks", *IEEE International Conference on Consumer Electronics 2025* (**ICCE'25**), Las Vegas, NV, USA, January 2025.
- "Enabling Practical and Pervasive Content Delivery from Emerging LEO Mega-Constellations", *IEEE International Conference on Multimedia and Expo 2024* (**ICME'24**), Niagra Falls, Canada, July 2024.
- "Achieving Predictable and Scalable Load Balancing Performance in LEO Mega-Constellations", *IEEE International Conference on Communications 2024* (**ICC'24**), Denver, CO, USA, June 2024.
- "Ares: Predictable Traffic Engineering under Controller Failures in SD-WANs", *ACM Web Conference 2024* (**WWW'24**), Singapore, May 2024.
- "RateSheriff: Multipath Flow-aware and Resource Efficient Rate Limiter Placement for Data Center Networks", *IEEE/ACM International Symposium on Quality of Service 2023* (**IWQoS'23**), Orlando, FL, USA, June 2023.
- "ProgrammabilityMedic: Predictable Path Programmability Recovery under Multiple Controller Failures in SD-WANs", *IEEE International Conference on Distributed Computing Systems 2021* (**ICDCS'21**), Virtual, July 2021.
- "Improving the Path Programmability for Software-Defined WANs under Multiple Controller Failures", *IEEE/ACM International Symposium on Quality of Service 2020* (**IWQoS'20**), Virtual, June 2020.

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